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The chromosomes of bacteria, viruses, and eukaryotic cells contain pieces of DNA that move around the genome. Such movement is called transposition. DNA segments that carry the genes required for this process and consequently move about chromosomes are transposable elements or transposons. Transposons are also called jumping genes because they can jump from one DNA to another, resulting in mutation of the cell. They were first discovered in 1951 by Barbara McClintock whose significant discovery was ignored by scientific community for many years. She was awarded the Nobel Prize in 1983.

12.8 Recombinant DNA Technology

One of the practical applications of microbial genetics and the technology arising from it is the recombinant DNA technology. The deliberate modification of an organism's genetic information by directly changing its nucleic acid genome is called genetic engineering and is accomplished by a collection of methods known as recombinant DNA technology. Recombinant DNA technology opens up totally new areas of research and applied biology. Thus, it is an essential part of biotechnology, which is now experiencing a stage of exceptionally rapid growth and development. In general sense, **recombination** is the process in which one or more nucleic acids molecules are rearranged or combined to produce a new nucleotide sequence. Usually genetic material from two parents is combined to produce a recombinant chromosome with a new, different genotype. Recombination results in a new arrangement of genes or parts of genes and normally is accompanied by a phenotypic change.

There are many diverse and complex techniques involved in gene manipulation. However, the basic principles of recombinant DNA technology are reasonably simple, and broadly involve the following stages (Figure 12.22).

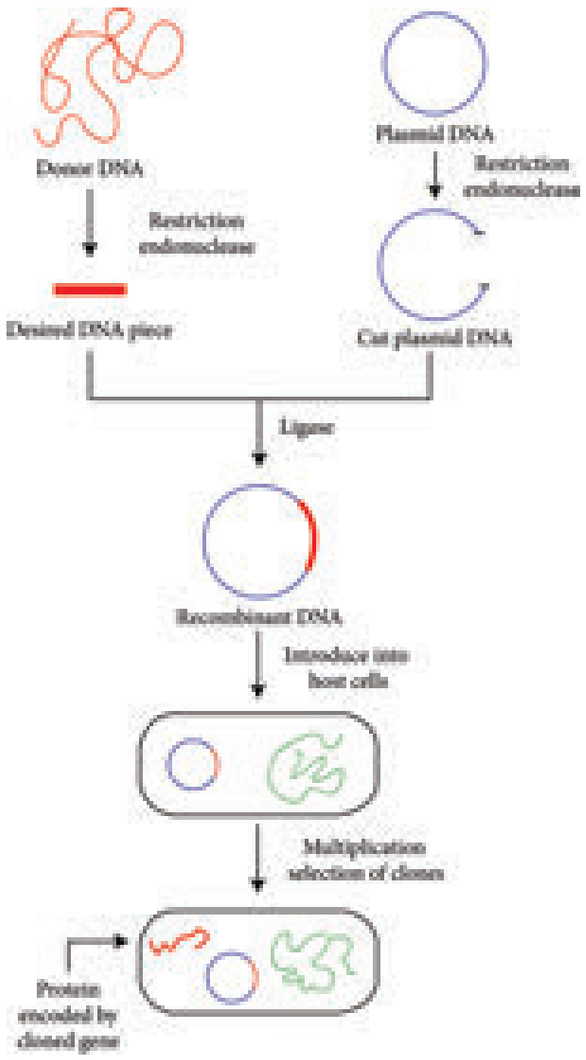


Figure 12.22: Basic principles of recombinant DNA technology

1. Isolation of DNA from the source (Donor)
2. Generation of DNA fragments and selection of the desired piece of DNA
3. Insertion of the selected DNA into a cloning vector (Example: a plasmid) to

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